ZIQI(OLIVIA) WANG

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Education

- **Rutgers University, New Brunswick, NJ, United States** *01/2025 - 12/2026(expected)*
  Master of Science in Computer Science , Department of Computer Science ***GPA**: 4.0/4.0*
- **Hebei University of Technology, Tianjin, China** *09/2020 - 06/2024*
  Bachelor of Literature in International Education of Chinese , School of Humanities and Law ***GPA**: 3.98/4.0*
- **2nd Prize in Graduation Scholarship,** 2023-2024 Academic Year, Hebei University of Technology *06/2024*

Publication

- **AURORA: Attention-intended Dual Verification with Adaptive Rollback for LLM Agents**
  ***Ziqi Wang**, Huaiyuan Yao, Hua Wei, Hang Zhou, Xiao Luo* *Under Review at IROS 2026*
- **RAGRouter-Bench: A Dataset and Benchmark for Adaptive RAG Routing** *(arXiv:2602.00296)*
  ***Ziqi Wang**, Xi Zhu, Shuhang Lin, Haochen Xue, Minghao Guo, Yongfeng Zhang* *Under Review at ARR 2026*

Research Experience

- **AURORA: LLM Agent Trajectory Verification** *12/2025 - 03/2026*
  Research Assistant, Supervisor: Xiao Luo, University of Wisconsin-Madison
  - **Dual-Head Verifier Design:** Designed an attention-based dual-head verifier using distinct masking strategies to produce local (step validity) and global (trajectory coherence) scores within a single shared encoder.
  - **Context Isolation Mechanism:** Developed AURORA-Isolation to mask trajectory history from the local head before attention, preventing representation interference and resolving score convergence under shared parameters.
  - **Adaptive Rollback & Evaluation:** Implemented adaptive rollback triggered by dual-dimensional verification signals, achieving +19.2pp success rate improvement over GA-Rollback baseline on ALFWorld across 3B–14B models.
- **RAGRouter-Bench: Adaptive RAG Routing Research** *05/2025 - 01/2026*
  Research Assistant, Supervisor: Yongfeng Zhang, Rutgers University
  - **Benchmark Construction:** Built RAGRouter-Bench, the first adaptive RAG routing benchmark covering 7,727 queries and 21,460 documents, standardizing 5 representative RAG paradigms under a unified retriever abstraction.
  - **Dual-View Routing Indicators:** Introduced dual-view query-corpus compatibility metrics (semantic + structural) to characterize RAG paradigm applicability across factual, reasoning, and summarization tasks.
  - **Evaluation Framework & Insights:** Designed a unified effectiveness-efficiency evaluation protocol, revealing that no single RAG paradigm universally dominates and corpus properties critically shape routing decisions.

Working Experience

- **Siemens Ltd., China** *05/2025 - 08/2025*
  LLM and Data Analysis Intern, Digital Industrial Software Dept. *Beijing, China*
  - **Fine-tuned Industrial LLM:** Fine-tuned an industrial-specific Qianwen2.5 32B model by combining LoRA and prompt engineering, reducing hallucinations and enhancing accuracy on technical terminology by 12%.
  - **Built RAG Knowledgebase:** Cleaned and structured extracted multimodal data with Pandas to build an industrial knowledge base in Elasticsearch with over 2,000 entities, providing the dataset for a downstream RAG system.
- **Turingops Co., Ltd.** *04/2024 - 11/2024*
  AI Engineer Intern, Product Development Dept. *Shanghai, China*
  - **Built Intent Classifier:** Co-developed a core user intent recognition module, using prompt engineering, embedding similarity, a fine-tuned classifier, and an LLM reflection mechanism to achieve 85% stable accuracy.
  - **Optimized Query Robustness:** Contributed to a query preprocessing module by using sub-question generation, query rewriting and structural decomposition to boost average answer accuracy on complex queries by 16%.
- **SOHU.com Limited** *01/2024 - 04/2024*
  Back-end Development Intern, Product Development Dept. *Wuhan, China*
  - **Back-end Developed:** Contributed to building microservices and RESTful APIs (Spring Boot) and optimizing database persistence layers, including implementing CRUD operations with JPA.
- **Danfoss Ltd., China** *03/2023 - 12/2023*
  Strategic Intern, Security Operation Center, Danfoss Group IT *Tianjin, China*
  - **Cybersecurity Maintenance and Research:** Developed automated Shell scripts for network traffic analysis and server diagnostics, supporting monitoring and failure analysis for a 1000+ device cluster and reducing troubleshooting time.

Teaching Experience

- **Teaching Assistant,** *CS501: Mathematical Foundations of Data Science, Rutgers University* *2026 Spring*
- **Teaching Assistant,** *CS462: Introduction to Deep Learning, Rutgers University* *2025 Fall*
- **Teaching Assistant,** *CS520: Introduction to AI, Rutgers University* *2025 Fall*

Skills

- **Computer Science:** *Programming Languages*: Python, C++, Java; *ML & DL*: PyTorch, TensorFlow, LoRA/PEFT, RAG; *Data*: Elasticsearch, Scrapy, Pandas, Docker
- **Languages:** Mandarin (Native), English (Proficient)